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Growth Charting of Brain Connectivity Networks and the Identification of Attention Impairment in Youth

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Abstract

Importance—Intrinsic connectivity networks (ICNs), important units of brain functional organization, demonstrate substantial maturation during youth. In addition, interrelationships between ICNs have been reliably implicated in attention performance. It is unknown whether alterations in ICN maturational profiles can reliably detect impaired attention functioning in youth.

Objective—To use a network growth charting approach to investigate the association between alterations in ICN maturation and attention performance.

Design, Setting, and Participants—Data were obtained from the publicly available Philadelphia Neurodevelopmental Cohort, a prospective, population-based sample of 9498 youths who underwent genomic testing, neurocognitive assessment, and neuroimaging. Data collection was conducted at an academic and children's hospital health care network between November 1, 2009, and November 30, 2011, and data analysis was conducted between February 1, 2015, and January 15, 2016.

Main Outcomes and Measures—Statistical associations between deviations from normative network growth were assessed as well as 2 main outcome measures: accuracy during the Penn Continuous Performance Test and diagnosis with attention-deficit/hyperactivity disorder.

Results—Of the 9498 individuals identified, 1000 youths aged 8 to 22 years underwent brain imaging. A sample of 519 youths who met quality control criteria entered analysis, of whom 25 (4.8%) met criteria for attention-deficit/hyperactivity disorder. The mean (SD) age of the youth was 15.7 (3.1) years, and 223 (43.0%) were male. Participants' patterns of deviations from

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normative maturational trajectories were indicative of sustained attention functioning (R^2 = 24%; $F_{6,512}$ = 26.89; $P < 2.2 \times 10^{-16}$). Moreover, these patterns were found to be a reliable biomarker of severe attention impairment (peak receiver operating characteristic curve measured by area under the curve, 79.3%). In particular, a down-shifted pattern of ICN maturation (shallow maturation), rather than a right-shifted pattern (lagged maturation), was implicated in reduced attention performance (Akaike information criterion relative likelihood, 3.22×10^{26}). Finally, parallel associations between ICN dysmaturation and diagnosis of attention-deficit/hyperactivity disorder were identified.

Conclusions and Relevance—Growth charting methods are widely used to assess the development of physical or other biometric characteristics, such as weight and head circumference. To date, this is the first demonstration that this method can be extended to development of functional brain networks to identify clinically relevant conditions, such as dysfunction of sustained attention.

Normative growth charts (eg, pediatric growth charts for height, weight, and head circumference) have been produced for physical or other biometric characteristics for at least 200 years and have proved valuable in the early identification of clinically relevant states and conditions.[1] With the advent of resting-state functional neuroimaging,[2] the potential now exists to extend growth charting methods to maturing functional relationships in the brain. The human brain is organized into several large-scale intrinsic connectivity networks (ICNs), each associated with distinct neurocognitive functions.[3,4] Moreover, relationships within and between ICNs exhibit clear trajectories of change from childhood to young adulthood.[5-9] It is plausible, then, that deviations from normative trajectories of network maturation might be predictive of a range of clinically important psychological characteristics and conditions.

Network growth charting methods might be particularly well suited to assessing sustained attention functioning and diagnosis of attention-deficit/hyperactivity disorder (ADHD). The capacity for sustained attention (ie, maintaining task-directed focus for extended periods of time) improves rapidly from early childhood to young adulthood.[10-14] In largely independent lines of research, influential new brain network models[15-22] propose that proper sustained attention functioning requires both (1) engagement of task-positive networks (TPNs), including frontoparietal control network (FPN) and dorsal and ventral attention networks (DAN and VAN), and (2) suppression of the default mode network (DMN),[23-27] an important ICN implicated in introspective attention and mind wandering. Individuals with ADHD exhibit reduced sustained attention[29-34] as well as distributed alterations in ICNs,[15,16] in particular, disrupted organization within the DMN and inadequate suppression of this network by TPNs.[19,35] Functional connectivity patterns within the DMN and between the DMN and TPNs exhibit massive maturation from childhood to young adulthood.[5-9] This observation raises an intriguing hypothesis that deviations from normative patterns of intranetwork and internetwork maturation, calculated by means of normative network growth charts, could reliably predict early emergence of sustained attention impairment and ADHD. To our knowledge, the present study is the first to directly investigate this approach.

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Methods

Sample

All data were obtained from the Philadelphia Neurodevelopmental Cohort,[36,37] a sample of 9498 youths distributed via the Database of Genotypes and Phenotypes dissemination platform.[38,39] After providing written informed consent (or assent with parental consent for participants younger than 18 years), participants underwent neurocognitive and genetic assessment. From this sample, 1000 participants underwent neuroimaging. Of these, youths who met the criteria for high-quality scans and motion correction were retained for analysis (below and eMethods in the Supplement). Demographic information is presented in the Table and in eTable 1 in the Supplement. The University of Michigan institutional review board determined that review was not required for the present study. Data collection was conducted at an academic and children's hospital health care network between November 1, 2009, and November 30, 2011, and data analysis was conducted between February 1, 2015, and January 15, 2016.

Task

Participants completed the Penn Continuous Performance Test consisting of 180 trials.[40] Stimuli are displayed once per second, and participants respond whenever the segments displayed form a digit or letter, depending on the task phase. We calculated accuracy as a proportion of correct trials out of all 180 trials (Table). Consistent with prior research,[10-14] accuracy during sustained attention improves substantially with increasing age ($F_{2,516}$ = 126.96; $P < 2.2 \times 10^{-16}$). Our goal was to build a regression model that predicts participants' task accuracy relative to what is expected for age. We thus corrected accuracy for participants' age by modeling task accuracy as a function of linear and quadratic age and retaining the residuals. This age-corrected accuracy variable was our primary dependent measure.

Clinical Interview

All participants underwent a computer-aided GOASSESS interview[41] (an abbreviated in-person interview modeled on the Schedule for Affective Disorders and Schizophrenia for School-Age Children–Present and Lifetime Version[42]). Parents or caretakers were interviewed to provide diagnostic information on youth aged 8 to 17 years, and those aged 11 or older were also interviewed directly.[43] The ADHD module has been shown to have excellent correspondence with the full Composite International Diagnostic Interview–Adolescent version.[43] We identified a subset of youths with ADHD based on (1) the presence of at least 2 symptoms of inattention (for inattentive type) or 2 symptoms of hyperactivity (for hyperactivity/impulsivity type) or both (for combined type), (2) the presence of symptoms in more than 1 context, (3) the presence of symptoms at the time of evaluation, and (4) the severity of these symptoms reported to be 5 or greater (0, not significant; 5, significant; 10, very severe). A total of 25 youths met these criteria (10, inactive type; 3, hyperactive/impulsive type; and 12, combined type), and their demographic information and medication status are reported in the Table.

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Preprocessing and Connectome Generation

Preprocessing and connectome generation were consistent with previous work.[9,44-47] Briefly, resting-state data were de-trended and nuisance effects were removed from each voxel's time course. Next, band-pass filtering was followed by motion scrubbing (where high-motion volumes are removed from the time series) using a 0.2-mm framewise threshold. We then calculated spatially averaged time series for each of 1068 regions of interest placed in a regular 12-mm grid throughout the brain. Finally, Pearson correlation coefficients, calculated between each region of interest, were transformed using Fisher *z*transformation. Resting-state functional connectivity quality-control plots[48,49] (eFigure 1 in the Supplement) showed virtually no residual systematic association between connection length and motion effects. For visualization, we used the network map of Yeo and colleagues[50] to assign regions of interest to large-scale ICNs.

Independent Components Analysis

We carried out a form of joint independent components analysis (ICA)[46,51-54] that we modified to be applied to a single modality while retaining the method's focus on characterizing intersubject variability. Additional details are provided in previous work[46] and in the eMethods in the Supplement. In brief, the steps are as follows. Each connection of the connectome underwent regression-based cleansing to remove nuisance variation. In addition, variance associated with effects of interest were augmented by adding the mean effect to induce selection of components related to effects of interest. After dimensionality reduction (standard in ICA) with model order set at 15 (chosen heuristically to correspond with previous work[46]), the FastICA algorithm was applied to these reduced data[55] to obtain component source maps and expression scores. Components were thresholded at $|z|>3$ for display. Stability was assessed with ICASSO,[56] which was run 1000 times and indicated that all components were stable (I_q ranged from 0.9366 to 0.9990).[57]

Network Growth Charting Analyses

Using our ICA-based method, we parsed the connectome into 15 cohesive components that we designated with the letters A through O. Each participant received an expression score for each component, which reflects the extent to which the component is expressed in that participant's connectome. We next created normative growth charts for each component by plotting a quadratic fit line for component expression against age. We then assigned a maturational deviation score for each component to each participant. These deviation scores reflect the differences in each participant's component expression relative to what is predicted by the person's biological age, with positive numbers reflecting overexpression relative to what is predicted for age and negative numbers representing underexpression.

Using multiple regression, we examined whether these maturational deviation scores were predictive of performance on the Penn Continuous Performance Test (ie, age-corrected accuracy) as well as ADHD status. Additional multiple regressions (as used above to generate fit lines) were used to assess whether specific components exhibited statistically significant maturation. In these regressions, component expression was predicted by linear and quadratic age terms and Bonferroni correction was applied for serial testing across multiple components. An overview of the overall analytic strategy is shown in Figure 1.

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Results

Network Growth Charting to Predict Task Accuracy

Using our network growth charting method, we found that the maturational deviation scores of the 15 connectomic components predicted age-corrected task accuracy ($F_{15,503} = 13.48$; $P < 2.2 \times 10^{-16}$) (eTable 2 in the Supplement reports regression coefficient test statistics). Moreover, a subset of 6 of the 15 components (components A-F) accounted for nearly all of the predictive power of the model ($F_{6,512} = 26.89$; $P < 2.2 \times 10^{-16}$; regression coefficients for all 6 components were statistically significant at $P < .001$). In particular, the full 15-component predictor model accounted for 28.7% of the variance (R^2) in age-corrected task accuracy, and the 6-component predictor model accounted for 24.0% of the variance. Of these 6 components, 5 (components A-E) showed clear evidence of vigorous maturation (tested using quadratic growth models; all $P < .05$, Bonferroni corrected for serial testing) (eTable 3 in the Supplement provides individual test statistics). For all 5 of these highly maturing components, downward deviations from the normative maturational trajectory (ie, underexpression relative to age) predicted worse accuracy.

To assess the replicability of our main analysis, a split-half analysis was conducted (eMethods in the Supplement) in which the participants were divided into a training sample and an independent test sample. This analysis showed that components learned on the training sample still had remarkable predictive power for age-corrected task accuracy when applied to the test sample ($F_{15,244} = 3.48$; $P = 2.1 \times 10^{-5}$; $R^2 = 17.6\%$), providing evidence of good replicability.

Shifting DMN-TPN Architecture Among Maturing Components

Each of the 5 strongly maturing components (ie, components A-E) implicated distinct cohesive changes across the connectome (Figure 2 and eFigures 2 and 3 in the Supplement), and each had distinct maturational curves (Figure 3 and eFigure 4 in the Supplement). Nonetheless, they shared many striking commonalities.

DMN Integration—Components A, D, and E exhibited increases in intra-DMN connectivity. For component A, intra-DMN integration (ie, increased connectivity with age) was widespread throughout the network. For components D and E, integration was relatively concentrated in certain nodes: angular gyrus for component D and posterior cingulate cortex and angular gyrus for component E.

Changing DMN-Attention Network Relationships—Components A, D, and E all showed changing interrelations between DMN and 2 major attention networks: DAN and VAN. In component A, there was extensive segregation (ie, decreased connectivity with age) between DMN and both DAN and VAN.

Changes in FPN and Its Relationships With Other Networks—The FPN was prominently represented in all 5 highly maturing components. Component B, which is both the strongest predictor of task performance as well as the most vigorously modulated by age, showed increasing connectivity within the FPN and DAN and increasing integration between

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these networks. More details about components A through F can be found in the eResults, eDiscussion, and eFigures 2-12 in the Supplement.

Shallow vs Lagged Dysmaturation and Task Accuracy

For components that exhibit an inverted U-shaped maturational trajectory (ie, B, C, and E), it is possible to distinguish 2 simple, distinct forms of altered maturation (eMethods and eFigure 13 in the Supplement). The first form consists of shallow maturational trajectories that are downshifted relative to the normative trajectory (in contrast, an upshifted curve reflects a more robust maturational trajectory). Alternatively, altered maturation could manifest as maturational lag, where a later onset of maturational change yields a right-shifted curve (in contrast, precocious individuals exhibit left-shifted curves). The lagged model showed a statistically significant fit ($F_{6,512} = 3.37$; $P = .0029$), but it accounted for relatively little variance ($R^2 = 4\%$). In contrast, the shallow model had excellent overall fit ($R^2 = 24\%$, as reported in the Results section). We quantitatively compared these models using the Akaike information criterion,[58] which showed overwhelming evidence in favor of the shallow model (relative likelihood of shallow vs lagged, 3.22×10^{26}). Relative likelihoods greater than 100 are considered decisive evidence.[59]

Biomarker of Attention Dysfunction From Network Growth Charting

We next assessed whether the maturational deviation scores for components A through F could serve as objective biomarkers of youth placed into dichotomous groups of normal performers and low performers on the Penn Continuous Performance Test. We defined *low performers* as youth who fell below the *x*th percentile in terms of age-corrected performance, where *x* was allowed to vary from a stringent 5th percentile to a liberal 50th percentile in 5% increments; *normal performers* were defined as the remaining youths. Logistic regression was used to classify youths as low or normal performers based on deviation scores. Leave-one-out cross-validation (eMethods in the Supplement) indicated that classification was reliable (Figure 4), with highly statistically significant findings for all cut points (peak accuracy was at the 10th percentile cut point, with 79.3% area under the receiver operating characteristic curve). For better appreciation of the nature of trade-offs in sensitivity and specificity that give rise to area under the curve values, we provide all receiver operating characteristic curves in eFigure 14 in the Supplement.

Biomarkers Of ADHD From Network Growth Charting

We used the maturational deviation scores for components A through F to carry out a logistic regression predicting ADHD diagnosis. The model as a whole was significant (likelihood ratio test, $\chi^2_6=13.00$; $P = .043$). In addition, components A through C were significantly predictive of diagnosis (odds ratios, 1.70 [$P = .0269$] for component A; 2.07 [$P = .0129$] for component B; and 1.87 [$P = .0160$] for component C) (eTable 4 in the Supplement reports all odds ratios and test statistics).

Discussion

A recent consensus review[60] challenges psychiatric neuroscientists who seek deeper understanding of neurodevelopmental conditions to “unravel the miswired

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connectome."[60](p1335) In our resting-state imaging study of 519 youth in the Philadelphia Neurodevelopmental Cohort, we used a novel network growth charting method. We mapped the normative maturational trajectories of major components of the functional connectome and showed that downshifted component expression relative to the normative profile (shallow maturation) is implicated in both impaired attention task performance and ADHD. Many psychiatric disorders are thought to have their origins in early neurodevelopmental events. Our results invite further investigation into the use of network growth charting to identify patterns of brain dysmaturation that can serve as early, objective markers of cognitive problems and disorder vulnerability.

Previous research has established that interrelationships between large-scale ICNs exhibit a highly reliable pattern of developmental change. Maturation of interrelationships between DMN and TPNs (in particular, FPN, DAN, and VAN) are among the most frequently reported in the literature,[5-9] and our results—especially in components A through E—are highly consistent with these previous findings. Separately, researchers have proposed that large-scale ICNs, especially DMN and TPNs (and their competitive balance),[15,16] play a critical role in sustained attention functioning. The frontoparietal control network plays a major role in adaptive attention control,[61,62] DAN subserves voluntary "top-down" attention based on goals, and VAN supports automatic "bottom-up" forms of attention.[63-66] Notably, abnormalities in all of these networks have been observed in ADHD.[16,67,68]

Our network growth charting method forges an important connection between the neurodevelopment of ICNs and their role in sustained attention. We showed that when individuals deviate from normative maturational trajectories in the development of ICNs, they consistently exhibit worse sustained attention performance. In addition, we achieved reliable dichotomous classification of youth as low and normal performers (based on sustained attention functioning during the Penn Continuous Performance Test), with reliability of classification generally increasing with the stringency of the cutoff used to define the low performance group. Our approach contrasts with that of Rosenberg and colleagues,[28] who used functional connectomes to predict sustained attention performance but who did not take a network maturation approach.

We also found that many of the same dysmaturation patterns that are predictive of poor attention performance are also predictive of ADHD, consistent with a previous study[9] using the ADHD-200 data set (encompassing 421 youth after exclusions; 135 with ADHD). There it was found that the neurotypical pattern of integration within the DMN and segregation of the DMN from attention networks is reduced in youth with ADHD, which is highly similar to our present results involving component A. Other nondevelopmental studies in youths[69,70] and adults[71-73] have also found alterations in DMN-attention network interconnections in ADHD.[67]

When testing which of 2 forms of dysmaturation (ie, shallow vs lagged) better explains worse attention performance, we found strong evidence in favor of the shallow model of dysmaturation. In this form of dysmaturation, there is reduced expression of the relevant component throughout the developmental trajectory (more precisely, throughout the age range investigated). Interestingly, previous studies of structural features of the brain (eg,

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cortical thickness[74] and cortical surface area[75]) have also shown that poor attention functioning, as indexed by ADHD diagnosis, is associated with shallow maturation. The aberrant maturation patterns observed across these structural and functional modalities may be related. This hypothesis gains support from the observation that prefrontal regions, especially superior and medial prefrontal cortex, were prominent in the preceding structural alterations that were found in ADHD, and these regions lie within the networks (especially FPN, VAN, and DMN) that are reliably implicated in our functional connectivity results. In addition, a previous multimodal investigation[46] found strong quantitative evidence that regions that exhibit alterations in a structural metric (gray matter volume) also tended to exhibit altered patterns of functional connectivity.

This study has several limitations. First, we used a machine learning–based data reduction method (joint ICA)[46,51-54] to identify components of the brain's functional architecture. All such methods use various assumptions (eg, components are nongaussian) and require setting numerous parameters (eg, number of components); future studies may find alternative settings that perform better. Second, although the ability of our regression models to predict sustained attention was impressive, it is important for researchers to replicate our findings in independent, comparable data sets, which can be achieved by at least 2 different routes: (1) components identified in the present study can be used to recover component expression scores from new datasets, which can be entered into predictive regressions or (2) new components and expression scores can be generated by applying our entire joint ICA pipeline to new data sets (eMethods in the Supplement). Third, the number of youths who met criteria for ADHD was small; thus, predictive results for ADHD should be interpreted cautiously. In addition, diagnosis of ADHD in the PNC data set was achieved with GOASSESS, an abbreviated clinical interview. Generalizability of our ADHD results to clinical samples diagnosed with standard clinical methods requires additional investigation.

Conclusions

This study introduces a novel brain network growth charting method for the prediction of attention impairment. Our results invite further investigation into the use of neuroimaging to identify patterns of brain dysmaturation that can serve as early, objective markers of cognitive problems and disorder vulnerability.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Key Points

Question

In the same way that pediatricians use growth charts of physical characteristics (height, head circumference) to identify somatic abnormalities, can we use growth charts of functional networks in the brain to identify neurocognitive abnormalities, in particular, impairments in attention functioning?

Findings

In this study of 519 youth (aged 8-22 years) who underwent resting-state scanning, deviations from normative patterns of brain network growth were significantly predictive of impaired sustained attention performance and diagnosis of attention-deficit/hyperactivity disorder.

Meaning

Growth charting of functional brain networks could provide a novel approach for constructing neuroimaging biomarkers for psychiatrically relevant conditions.

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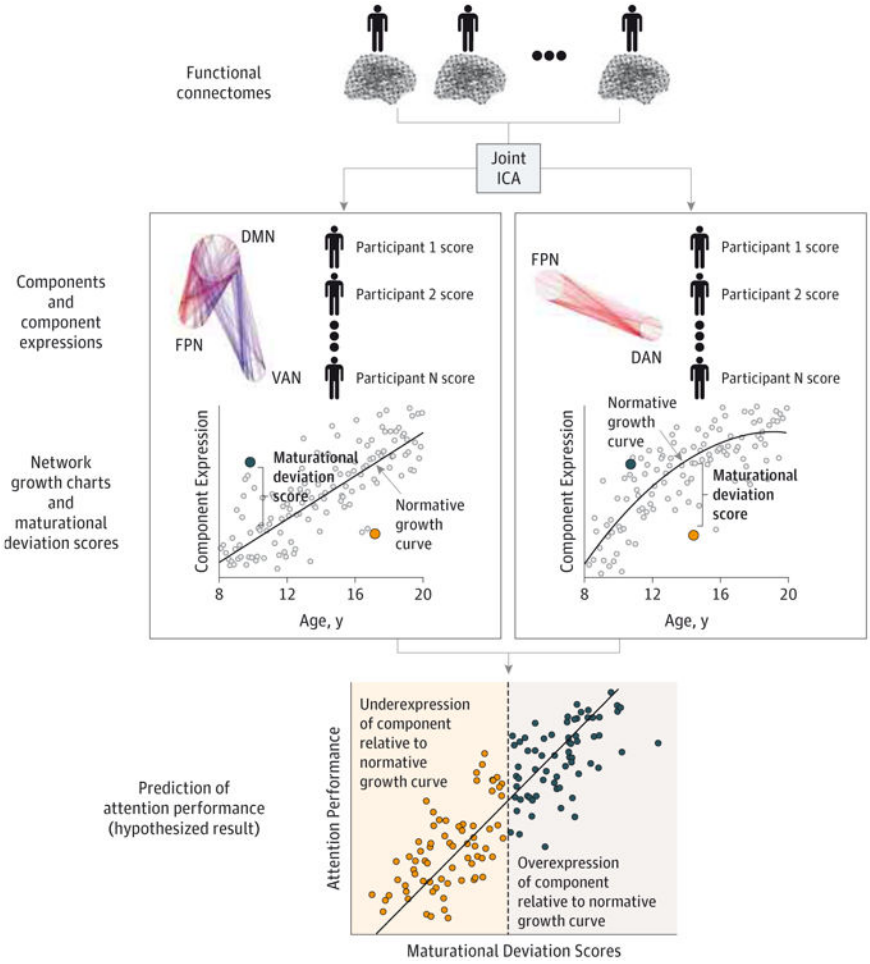
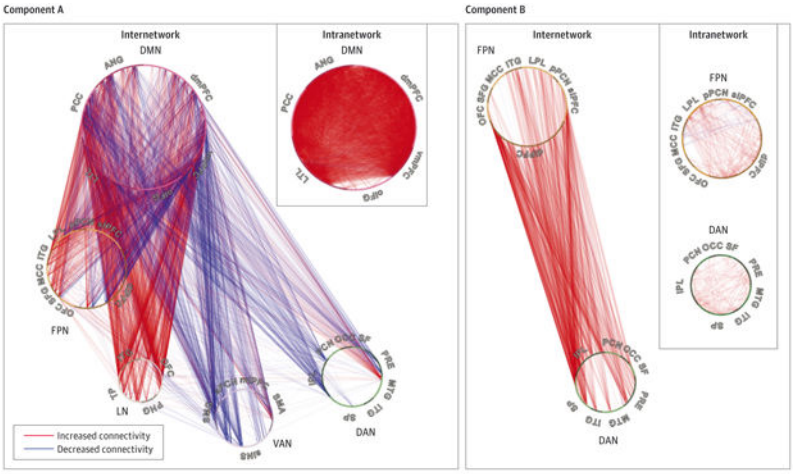


Figure 1. Schematic Diagram Illustrating Key Analysis Steps

Resting-state functional connectomes from 519 participants enter joint independent components analysis (ICA), which parses the connectomes into several cohesive components (2 illustrative components are shown in the boxes at middle). For each component, growth charts are then constructed that depict the normative change in component expression with age. Next, each participant is assigned a maturational deviation score for each component that reflects the degree to which that component is underexpressed or overexpressed relative to what is expected by age. These maturational deviation scores are then used as predictors of performance on a sustained attention task. DMN indicates default mode network; FPN, frontoparietal control network; VAN, ventral attention network.

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Abbreviation	Region Name
VN	**Visual Network**
FUS	Fusiform gyrus
LING	Lingual gyrus
CNS	Cuneus
SOC	Superior occipital cortex
MOC	Middle occipital cortex
IOC	Inferior occipital cortex
SMN	**Somatomotor Network**
SMA	Supplementary motor area
PRE	Precentral gyrus
POST	Postcentral gyrus
STG	Superior temporal gyrus
DAN	**Dorsal Attention Network**
SF	Superior frontal
PRE	Precentral gyrus
MTG	Middle temporal gyrus
ITG	Inferior temporal gyrus
SP	Superior parietal
IPL	Inferior parietal lobule
PCN	Precuneus
OCC	Occipital cortex
VAN	**Ventral Attention Network**
mIPFC	Middle lateral prefrontal cortex
SMA	Supplementary motor area
aINS	Anterior insula
SMG	Supramarginal gyrus
aPCN	Anterior precuneus

Abbreviation	Region Name
LN	**Limbic Network**
OFC	Orbitofrontal cortex
PHG	Parahippocampal gyrus
TP	Temporal pole
ITG	Inferior temporal gyrus
FPN	**Frontoparietal Network**
sIPFC	Superior lateral prefrontal cortex
dlPFC	Dorsolateral prefrontal cortex
OFC	Orbital frontal cortex
SFG	Superior frontal gyrus
MCC	Mid cingulate cortex
ITG	Inferior temporal gyrus
LPL	Lateral parietal lobule
pPCN	Posterior precuneus
DMN	**Default Mode Network**
dmPFC	Dorsomedial prefrontal cortex
vmPFC	Ventromedial prefrontal cortex
oIFG	Orbital inferior frontal gyrus
LTL	Lateral temporal lobe
PCC	Posterior cingulate cortex
ANG	Angular gyrus

Figure 2. Connectomic Maps for Components A, B, and C

Large circles represent individual intrinsic connectivity networks (based on the network parcellation by Yeo and colleagues[50]) and each dot within a large circle represents a region of interest (ROI) within that network. To further aid interpretability, groups of ROIs are assigned to anatomical regions within a network. Each line reflects a superthreshold connection; red lines reflect increased connectivity while blue lines reflect decreased connectivity. These highly maturing components consistently show prominent alterations within and between default mode network and task-positive networks (TPNs), including dorsal attention network (DAN), ventral attention network (VAN), and frontoparietal control network (FPN) (eMethods in the Supplement provides details on component display).

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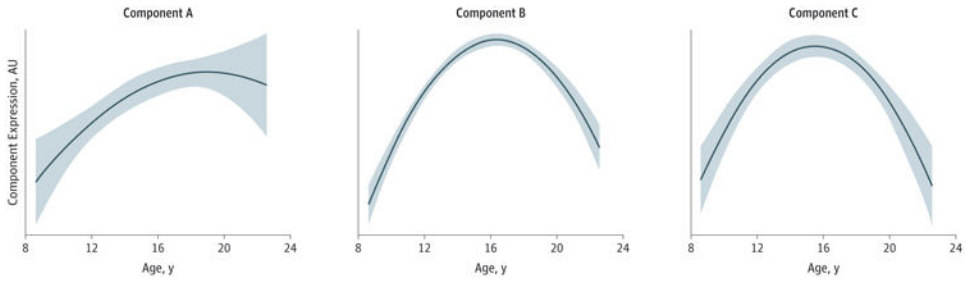


Figure 3. Maturation of Connectomic Components

Maturational profiles of components A, B, and C, 3 highly maturing components. The solid line is the quadratic fit, and gray shading represents 95% CIs for a quadratic model. Component expression is in arbitrary units (AU).

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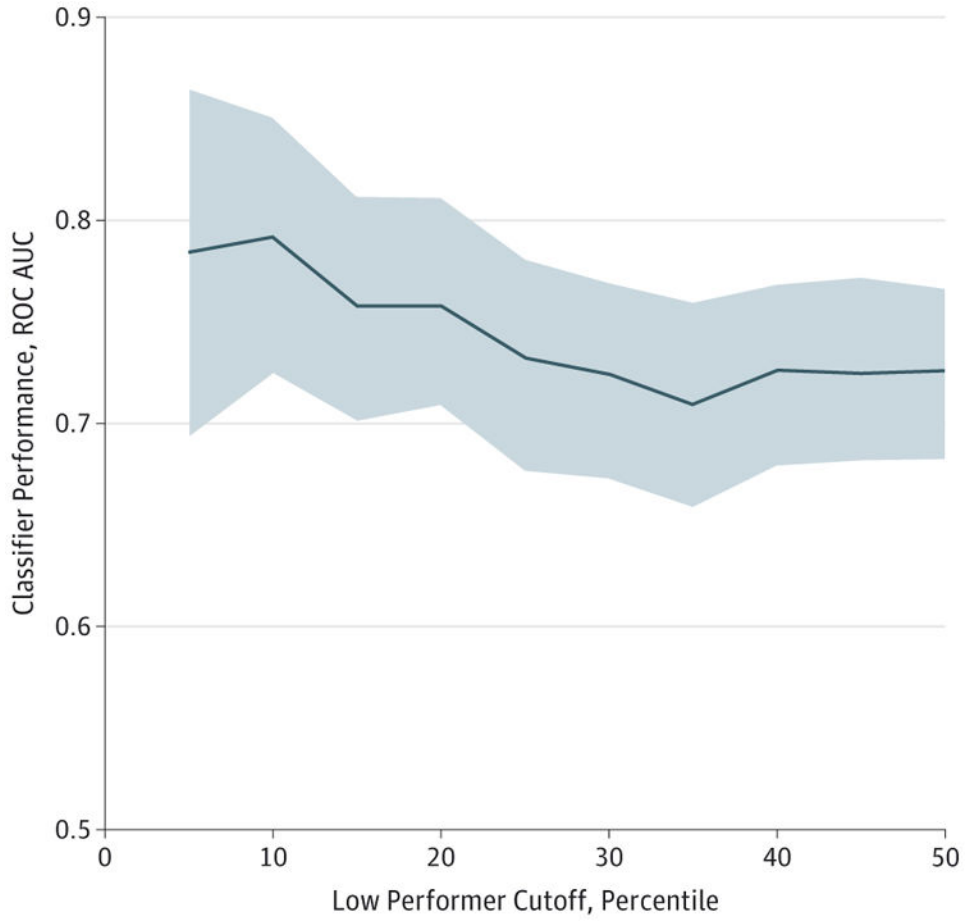


Figure 4. Performance of Logistic Regression Classifiers in Distinguishing Lowvs Normal Performers in a Sustained Attention Task

For each connectomic component, we calculated maturational deviation scores that quantify the extent to which expression of the component deviates from age-typical levels. We entered maturational deviation scores for components A to F into a logistic regression model to classify participants as low and normal performers on a sustained attention task. The percentile used to define low performers ranged from a stringent 5th percentile to a liberal 50th percentile (with the remaining participants considered normal performers). The performance of the classifier in a leave-one-out cross-validation framework was measured by area under the curve (AUC) for receiver operating characteristic (ROC) curves. Classifier performance was significantly better than chance for all performance cut points tested and reached a high of 79.3% in identifying youth in the bottom 10% of performance vs the remaining participants. Gray shading indicates 95% CIs; the solid line represents classification performance as measured by AUC ROC.

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Table

Summary of Sample and Key Measures[a]

Characteristic	ADHD (n = 25)	No ADHD (n = 494)	Total (N = 519)
Age, mean (SD), y	14.8 (4.2)	15.7 (3.1)	15.7 (3.1)
PCPT accuracy, mean (SD), %	85.29 (10.53)	90.04 (6.59)	89.82 (6.89)
Race/ethnicity, No. (%) [% Hispanic]			
White			
Male	4 (16.0)	114 (23.1) [1.8]	118 (22.7) [1.69]
Female	2 (8.0)	124 (25.2) [1.6]	126 (24.3) [1.59]
African American			
Male	10 (40.0)	69 (14.0)	79 (15.2)
Female	5 (20.0)	134 (27.1) [3.0]	139 (26.8) [3.6]
Asian			
Male	0	0	0
Female	0	3 (0.6)	3 (0.4)
American Indian/Native Alaskan			
Male	0	0	0
Female	0	2 (0.4)	2 (0.4)
Mixed race/other			
Male	2 (8.0)	24 (4.9) [37.5]	26 (5.8) [34.6]
Female	2 (8.0)	24 (4.9) [29.2]	26 (5.8) [26.9]
Medication use, No. (%)			
Yes	8 (32.0)	39 (8.0)	47 (9.1)
No	17 (68.0)	433 (87.7)	450 (86.7)
Unreported	0	22 (4.5)	22 (4.2)

Abbreviations: ADHD, attention-deficit/hyperactivity disorder; PCPT, Penn Continuous Performance Test.

[a]Extended demographic and in-scanner motion data are reported in eTable 1 in the Supplement.